MIL-C-48513(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

CASE SECTION, BALLISTIC, REAR

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of ballistic rear section entitled Case Section, Ballistic, Rear which is used on the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-48512	- Shipping and Storage Container, Atomic Warhead Section: M480 and M480E1.
MIL-T-5021	- Tests; Aircraft Welding Operators Certification.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8822047	- Box, Packing, Ammunition, For Case, Section, Ballistic, Rear.
8825074	- Case Section, Ballistic, Rear.
8825114	- Pressure Diaphragm Assembly.
8825120	- Frame, Nose Base, Aft.
8825121	- Frame, Nose Base, Aft.
8825153	- Adaption Kit, Atomic Warhead to Rocket, M86.

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8826386 - Case, Atomic Warhead Section.
8831129 - Mark 31 Warhead Compatibility Information.
EL-8842559 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The case section shall comply with all requirements specified on the drawing (dwg) 8825074, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Mating assembly.

3.5.1 Adaption kit assembly. The case section, ballistic, rear shall be capable of mating mechanically with the applicable mating assemblies shown on dwg 8825153 or their equivalents when tested as specified in 4.4.1.

3.5.2 Warhead compatibility. The case section, ballistic, rear shall be capable of mating mechanically with the applicable mating assembly shown on dwg 8831129 when tested as specified in 4.4.2.

3.6 Waterproofness. The assembly shall not allow visible leakage inside the assembly when subjected to a water spray simulating heavy rain for a period of 15 minutes as specified in 4.4.3.

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3.7 Environmental testing. The case section shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.7.1 Temperature shock. The case section shall withstand temperature shock between minus (-) 65° plus or minus (±) 5°F and 160 ± 5°F without evidence of damage to protective finishes, caulking material, or any permanent warpage after being tested as specified in 4.4.4.1.

3.7.2 Humidity. The case section shall withstand exposure to a relative humidity of 95 ± 5 percent at temperatures cycling from 80 ± 20°F to 160 ± 5°F to 80 ± 20°F without evidence of flaking or impairment of adhesion of the protective finish or caulking, deterioration or other damage when tested as specified in 4.4.4.2.

3.7.3 Low temperature cycling. The case section shall withstand, without damage, low temperature and frost conditions when tested as specified in 4.4.4.3. Upon completion of this test the assembly shall comply with 3.8.

3.7.4 Handling shock and vibration. The crated case section shall withstand without damage, the applicable requirements of the shock test of MIL-S- when tested as specified in 4.4.4.4. Upon completion of this test the assembly shall comply with the requirements of 3.5.1 and 3.6.

3.7.5 Flight vibration. The case section shall withstand without damage, flight vibration along the geometric (flight) axis over a frequency range of 75-1000-75 Hertz (Hz) at a peak gravity (g) level of ± 5 g's (± 10 percent tolerance at 80 ± 20°F). The vibration shall be applied to the assembly from low to high and back to low frequency over a period of 0.9 ± 0.1 minute. The assembly shall comply with the requirements of 3.8 after being tested as specified in 4.4.4.5.

3.8 Deterioration. Any internal or external components which show damage, evidence of deterioration or corrosion shall provide reason to consider the assembly as having failed to meet the test to which it was subjected.

3.9 Welding. Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of good practice on the part of the operator. All welding fluxes, scale, weld spatter, guide, and basic solutions shall be removed prior to application of any finish coat.

3.9.1 Welding operators. All fusion welding shall be performed by operators who are currently certified as class "B" welders in accordance with MIL-T-5021.

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3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 8825074) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Adaption kit assembly (see 3.5.1)		X
3. Warhead compatibility (see 3.5.2)		X
4. Waterproofness (see 3.6)		X

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5. Temperature shock (see 3.7.1)	X	
6. *Humidity (see 3.7.2)	X	
7. *Low temperature cycling (see 3.7.3)		X
8. Handling shock and vibration (see 3.7.4)		X
9. Flight vibration (see 3.7.5)		X

*Only 2 of first article assemblies shall be tested.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Pressure diaphragm assembly (see dwg 8825114 covering a detail of dwg 8825074).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Improper welding (see 3.9)	Visual
Major:		
	101. Improper hole location	Gage
	102. Outside diameter	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.2 Frame, nose base, aft (see dwgs 8825120 and 8825121 covering a detail of dwg 8825074).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper 27.607-0.016 diameter	Gage
	102. Improper 27.539-0.016 diameter	Gage
	103. Improper 29.644-0.020 diameter	Gage
	104. Improper 29.987-0.020 diameter	Gage
	105. Improper 29.870-0.020 diameter	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.3 Case section, ballistic, rear (see dwg 8825074).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Eccentricity at stations 114.75 and 99.0	TIR
2.	Concentricity of 27.010 ± 0.010 inch diameter on ring adapter with functional centerling on the 25.400 inch diameter bolt circle	TIR
3.	Parallelism of forward face of ring adapter with surface at station 115	Gage
4.	Mismatch at joint of frame nose base and nose base skins	Gage
5.	Gap at joints of frame nose base and nose base skins	Gage
6.	Gap at longitudinal joint of nose base skins	Gage
Major:		
101.	Mislocation of warhead mounting hole location	Gage
102.	Mislocation of mounting holes on 25.400 inch bolt circle	Gage
103.	All screws, lockbolts, and rivets not flush (allowed 0.015 inches below skin surface)	Gage
104.	Mislocation of case section, ballistic aft attach holes	Gage
105.	Evidence of poor workmanship (see 3.10)	Visual
106.	Incorrect or illegible serial number (see 3.4)	Visual
107.	Diaphragm is improperly orientated	Visual
108.	Any of the parts missing, damaged or loose	Visual/Manual
109.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Dents (0.05 inch maximum depth, concave, with a maximum diameter of 3 inches, 0.030 inch maximum rise; convex with a maximum diameter of 2 inches, no limitation of spacing)	Gage

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202. Scratches (0.010 to 0.030 inch depth shall not exceed 10 inches in length, scratches of less than 0.010 inch in depth no limitation in length; no limitation of width.....Gage
203. Improper paint coverage.....Visual
204. Improper paint color.....Visual
205. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.4 Box, packing, ammunition for case section, ballistic rear (see dwg 8822047).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Bag not sealed	Visual
	102. Marking missing, incorrect or mislocated	Visual
	103. Strapping missing, loose or damaged	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Adaption kit assembly	Major
b. Warhead compatibility	Major
c. Flight vibration	Major

Of these tests, c is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Adaption kit assembly, warhead compatibility, and flight vibration. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.4.5. If any item fails to comply 3.5.1, 3.5.2 and 3.7.5, the lot shall be rejected.

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Table IV

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8842559). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Adaption kit assembly. The case section, ballistic rear assembly shall be mounted to a fixture equivalent to the mating surface on Drawing 8825153.

4.4.2 Warhead compatibility. When testing for warhead assembly requirements, a functional cargo, described in Drawing 8831129 Mark 31 Warhead Compatibility Information or inert equivalent shall be used.

4.4.3 Waterproofness. The case section shall be assembled as part of the 8825153 assembly or equivalent, and the assembly shall be mounted horizontally and impacted with a water spray directed within 15 degrees of vertical. The rain test chamber shall be maintained at 80 ± 20°F throughout the test period. A simulated rain fall of 10 ± 2 inches per hour shall be produced by means of a water spray nozzle. The rain fall shall be rotated once about the longitudinal axis. Following the exposure period the assembly shall be removed from the test

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chamber, wiped dry externally, disassembled from the rest of the 8825153 assembly and checked for water leakage.

4.4.4 Environmental testing (see 3.7).

4.4.4.1 Temperature shock. The case section shall be placed within a test chamber where the stabilized temperature (see 6.4) is -65 + 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum at the conclusion of which, and within 5 minutes, the assembly shall be transferred to a chamber having an internal temperature of 160 + 5°F. The item shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of complete cycles shall be three with a maximum of 5 minutes transfer time between test chambers when going from one cycle to another. At the conclusion of the third cycle the assembly shall be removed from the test chamber.

4.4.4.2 Humidity. The case section shall be assembled as part of the 8825153 assembly or equivalent and the assembly shall be placed in the test chamber mounted to simulate the installed conditions. The chamber stabilized temperature shall be 80 + 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 + 5°F. The temperature of 160 + 5°F shall be maintained during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 80 + 20°F which completes one cycle. The relative humidity throughout the cycle shall be 95 + 5 percent. The test shall consist of 15 such cycles (360) hours.

4.4.4.3 Low temperature cycle. The case section shall be placed in the test chamber and set up to simulate installed conditions. The chamber stabilized temperature shall be 80 + 20°F. During the first 2 hour period the temperature shall be gradually reduced to -65 + 20°F. The temperature of -65 + 20°F shall be maintained for 4 hours. During the following 2 hour period the temperature shall be gradually increased to 80 + 20°F. The test chamber shall be maintained in this condition for 4 hours. This completes one cycle. The case section shall be subject to ten cycles.

4.4.4.4 Handling shock and vibration. The case section shall be mounted in the shipping container, case, atomic warhead section 8826386 or equivalent. The mounted assembly shall be tested in accordance with the shock test of MIL-S-

Following these tests the assembly shall be tested as specified in 4.4.1 and 4.4.3 of this military specification.

4.4.4.5 Flight vibration. The case section shall be mounted to a vibration producing apparatus at the mounting surface at station 115. The assembly shall then be vibrated along the geometric (flight) axis of the assembly at the frequencies, accelerations and the durations specified in 3.7.5. The acceleration specified shall be determined by suitable measuring device(s) attached to the mounting surface at station 115, at the point of attachment. At the conclusion of this test the assembly shall be inspected as specified in 4.3.2.1.1.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The case section shall be packaged in accordance with dwg 8822047 using the specified moisture barrier.

5.1.2 Level C. The case section shall be packaged in accordance with dwg 8822047 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The cable section shall be packed in accordance with dwg 8822047.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822047.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1677 dated 29 August 1961.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1115-A046

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